

Morphology-Conditioned World Model for Cross-Embodiment Quadrupedal Locomotion

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Problem & Setup



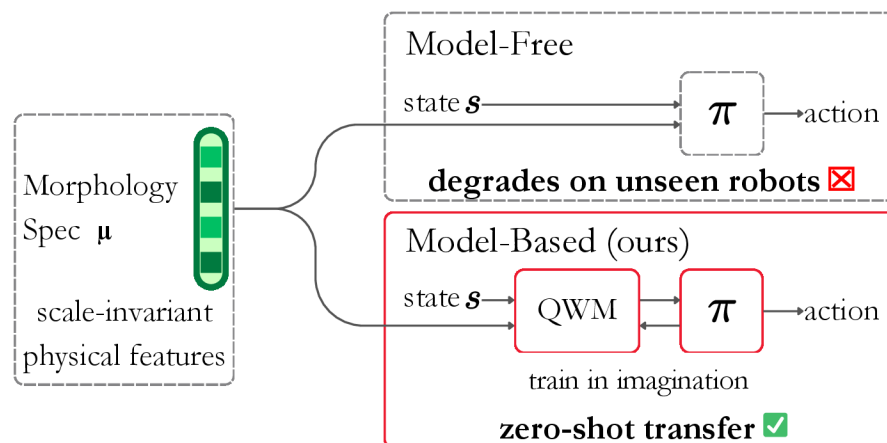
8 quadrupeds, 12 DoF each: different sizes (~6 kg → 50+ kg), knee types (X vs. dog-like), and actuators

World models are morphology-locked:

- A dynamics model trained on one robot fails on another.
- A new morphology means retraining from scratch.

Goal: one world model for a whole family of robots.

Same μ , two routes. Where should μ go?



Can a world model control a robot it has never seen?

Yes, condition it on the robot's morphology.

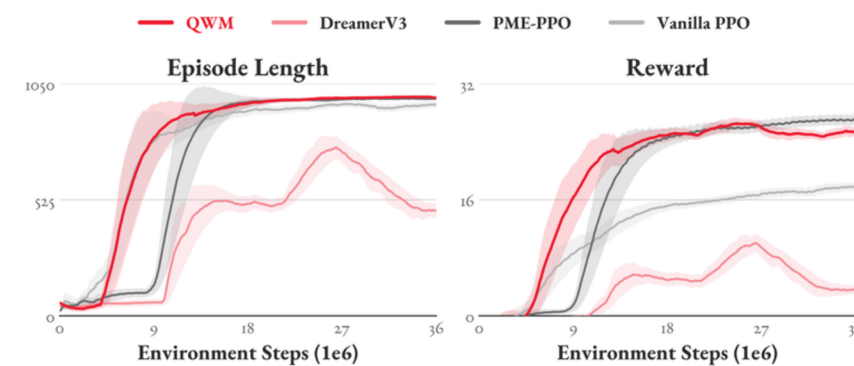
One world model. Unseen quadruped. Zero-shot.

First world model with zero-shot cross-embodiment transfer across quadrupeds.
Dynamics vary smoothly with morphology. Optimal control doesn't.



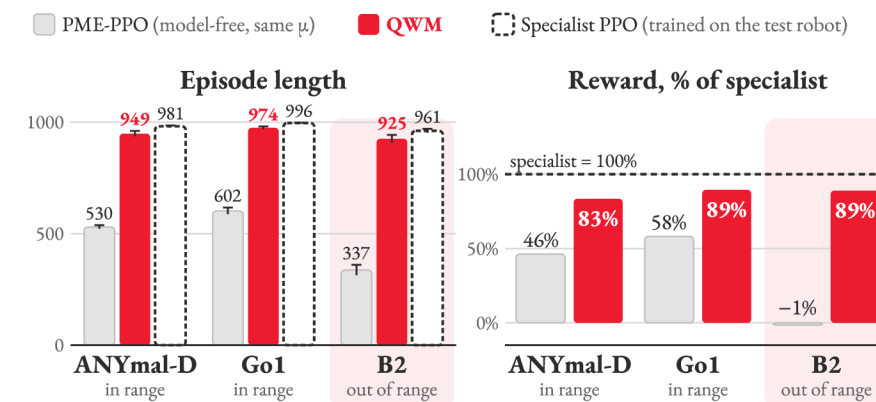
Results

On training robots, QWM matches PME-PPO with half the environment steps.



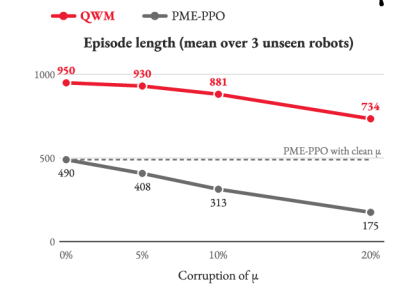
QWM and PME-PPO, both given μ , reach the same final performance. Without μ , Vanilla PPO plateaus lower and DreamerV3 fails to learn.
So any gap on unseen robots is generalization, not capability.

On unseen robots, QWM nearly matches a policy trained specifically on that robot. PME-PPO doesn't.



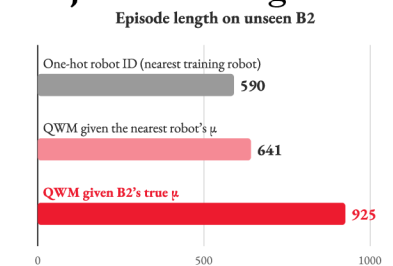
in range: body within the training robots' range
 out of range: B2 is bigger, heavier and stronger than any training robot.
 On B2, QWM keeps 96% of the specialist's episode length and 89% of its reward.
 PME-PPO, given the same μ , keeps 35% and about 0%.

What if μ is wrong?



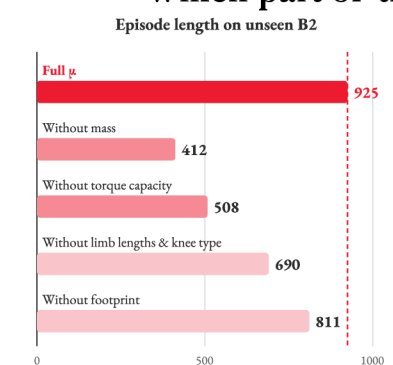
QWM with 20% corrupted μ still beats PME-PPO with a perfect μ , on all three robots.

Is it just matching B2 to the nearest training robot?



No. The physical values in μ carry the transfer.

Which part of the body matters?



Mass and torque capacity carry most of the transfer.

Is every piece needed?

Full QWM Fastest learning, lowest variance, best final reward	
No reward normalizer (ARN) Nothing learns: reward stays near 0 Reward scales differ 14x: Spot ~350 vs ANYmal-D ~25 per episode	No μ at all (no PME) Full-length episodes, but lower reward
No μ in the encoder Learns at first, then collapses late in training	No μ in the recurrent state Learns slower and ends at a lower reward